Renewable and Sustainable Energy Reviews 219 (2025) 115865

Contents lists available at ScienceDirect

Renewable and Sustainable Energy Reviews

journal homepage: www.elsevier.com/locate/rser

Market zones and electrification – A case study on the impact of market zone configuration on power-to-heat

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ARTICLE INFO

Keywords:
Electricity markets
Market zones
Nodal markets
Impact on system design
Impact on system operation

ABSTRACT

With the expansion of variable renewable electricity sources (VRES), electrification of heating, transportation and industry, and increased electricity integration of energy systems across country borders, electricity market structures and market behaviour impact not only investment and operation decisions in individual country systems but also across borders. This article explores this based on the case of Germany which has a single market-zone electricity market despite bottlenecks between north and south, resulting in the purchase of curtailment north of the country in Denmark during periods of high VRES-based production. This provides an income for Danish wind turbine operators, but it also means that prices are artificially high in Denmark during these hours resulting in lost opportunities for Power2x and Power2Heat operation. It ultimately impacts the feasibility of some of the main elements of the Danish energy transition which is based on the electrification of various sectors. For the consumer, it also raises concern regarding why wind turbines curtail during hours of high prices. This article addresses these issues in the German and Danish context and explores effects on the energy system. A market reconfiguration with more market zones in Germany would lower electricity prices in Denmark thus improving the feasibility of both electricity-intensive processes and electrification efforts while also benefitting Danish electricity consumers in general. It will, however, not necessarily benefit renewable energy-based producers.

1. Introduction

The need for an energy transition is well-recognised, thus most countries in Europe are in the process of expanding the exploitation of renewable energy sources (RES) – and even more explicitly, VRES. This is due to the circumstance that dispatchable RES are in short supply, with limited potential for expansion of stored hydropower [1] and resource and environmental restrictions on biomass resources [2,3]. At the same time, there is a push for and research interest in the direct or indirect electrification of industry [4], transportation [5–8], and heating [9–13], which combines well with an expansion of VRES – notably wind power and photo voltaics (PV). This electrification could optimally be assisted by the tendency of, e.g., wind power to drive down electricity costs [14–16] and the overall needs of sector-integrated smart energy systems [17] to provide flexibility.

Electricity systems across Europe are run according to spot markets, where suppliers and buyers typically are bidding into day-ahead markets where optimal prices are identified in the individual hours. This is typically done in marginal cost auctions, where the most expensive bid-taker in the given hour sets the clearing price [18]. Due to grid constraints, however, there are limits to how much power that can be transmitted between different countries. If this is the case, markets split up, and the implication is, that countries can end up with an electricity price which diverges from the price in neighbouring countries in a given hour. Not only this, even inside countries can there be a potential market split. Norway, for instance is split up into five market zones, Sweden is split up into four market zones, and Denmark is split up into two and depending on transmission capacity availability circumstances in the given hour, price levels can align or diverge despite being traded in the same spot market.

In contrast to this, e.g., Finland, Romania, Germany, Poland, Great Britain and France all form just one single prince zone despite considerable size and potential internal transmission grid constraints. Germany has been affected by opposition to transmission grid expansion

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https://doi.org/10.1016/j.rser.2025.115865
Received 2 October 2024; Received in revised form 13 May 2025; Accepted 14 May 2025
Available online 22 May 2025
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[19], but despite this situation and the circumstance that power cannot flow freely due to internal bottlenecks in the transmission grid, it remains a single-price market zone, with consequences for other countries connected to Germany.

While there are requirements for electricity market designs in Europe, they are also by nature political constructs. Thus, existing market designs and notably zonal configurations reflect to some extent political choices – and changes will equally reflect a choice. In this article, we explore some of the consequences of the political choices reflected in market designs.

Since the establishment of the pan-European electricity market there have been developments in the composition of the markets, and there is ongoing research into future developments and the effects of such. Hurta et al., for instance, analysed the impacts of a 2018 split of a common Austrian-Germany prize zone into two [20]; a move that was largely put in motion to address grid constraints between the two countries. Deilen et al. also describe how zonal reconfiguration is an issue in Central Western Europe [21], and Halbrügge and coauthors [22] even talk about a European system evolving with a *"split or aggregation of bidding zones"* as well as with changes in interconnectors between these zones.

In the Danish, Swedish, Finish and Norwegian power system, market coupling and splitting extends back to 1993 [23], where market coupling refers to adjoining areas establishing a common platform for the trade of power.

Analyses have been conducted regarding a splitting of the Germany electricity market into more zones. Ref. [24] investigated such a reconfiguration, noting it would generate higher prices in South Germany, thus providing incentives for power production expansion there or demand migration to the lower price areas. In the longer run, according to their analyses, the market zones would converge in terms of price level – though at a level higher than the existing single price market zone level. While providing incentives for capacity expansion in the south of Germany, it would clearly also affect German industry in the south negatively. This is supported by an older study where Kristiansen [25] investigated the power trading between the two market zones in Denmark and Germany, finding a general trend of higher prices in Germany than in Western Denmark, driving up prices in Denmark.

Egerer et al. [26] also analysed the effect of splitting Germany into two market zones – a northern and a southern. They focused on the effects in Germany and the changes in dispatch in two zones that such a split would cause. The split would result in lower prices in the northern market zone and higher in the southern zone. Fraunholz et al. [29] also looked into the effects of splitting Germany into two market zones, finding that *"Market [zone] splitting strongly affects day-ahead prices, expansion plans, [and] grid congestion"*. The electricity prices in the day-ahead market diverge considerably in their analyses, thus establishing different incentives in different ends of the country.

Similarly, Trepper et al. [27] analysed a market zone split in Germany motivated by increasing redispatch issues, finding that such a split would reduce redispatch measures. Redispatch refers to the situation where traded volumes cannot be effectuated due to grid constraints ("congestion"), and measures are taken to solve imbalances. They also split Germany into two market zones – a northern and a southern – based on a more detailed look into bottlenecks in the German transmission system. The authors also investigated the effects on other European countries, finding that prices in, e.g., the Netherlands, Denmark, Norway, Sweden, and Finland would decrease. Titz [28] found that wind power in Germany and trade with Denmark were main drivers for redispatch, and that their results supported a zonal split in Germany.

Short- and long-term impacts of splitting the German electricity market into two zones has also been analysed by Fraunholz and coauthors by combining an agent-based market model with an optimal power flow model in a joint modelling framework for the German system [97]. Their analysis show that the impacts are highest in the initial years starting from 2020, with significant divergence in the prices between the northern- and southern-German market zones. In the medium-term (2035) the price differences decline because of incentivized transmission expansion between the zones and allocating investments in generation capacity to the south zone. In the long-term (2050) the study shows that price differences increase again due to expansions of VRES capacity and lack of additional grid extensions. From a social welfare perspective, they find that producer benefits increase in the southern-German market zone, which overcompensate for consumer disadvantages in the zone, whereas the opposite pattern is found in the northern zone with decreasing producer rents and increasing consumer benefits. Overall, they conclude a negative welfare effect in Germany as a result of the zonal split – but highlights that including neighbouring electricity markets may affect this conclusion and the authors suggests policymakers and regulators to reassess the current single market zone approach.

Plancke et al. [30] analysed the implications of reconfiguring the German single market zone into two – a northern and a southern – and find that such solution results in less need for redispatch actions and a more efficient congestion management. The study further shows that this reconfiguration will cascade through other European markets—boosting consumer surplus in some areas while reducing producer surplus, and vice versa. The results show that consumers in Northern Germany, Scandinavia and Poland benefit in a social welfare perspective from splitting the German market zone into two, whereas producers in Southern Germany, France, Austria, the Czech Republic among others will benefit.

Klie and Madlener [31] suggest a nodal-pricing Contracts for Difference (CFD) auction for wind power expansions in Germany, in which developers bid their Levelised Cost of Energy (LCOE) as the strike price, but actual remuneration is reduced by each project's hourly nodal market value (accounting for congestion, curtailment and redispatch). Projects are then ranked by the resulting gap between the LCOE and the specific market value and awarded in ascending order. The authors argue that this ensures that locations alleviating grid bottlenecks while delivering high-value output are prioritised thus handling suboptimal siting practices of zonal markets.

Grimm et al. [32] looked into the discrepancy between investments in generating equipment which is largely a business economic consideration and investments in transition capacity, which is typically a Transmission System Operator (TSO) responsibility and thus under some level of societal oversight. They found that a German market zonal split would impact the location of generating equipment investment – and would lower transmission grid expansion. Similarly, Fink, analysing Sweden, found a tendency that PV investments were shifted to high-price market zones [33].

In another study on Sweden [34], Loiacono and coauthors analysed the electricity market, stressing transmission constraints as a problem for large electricity market zones, mentioning how Sweden in 2011 was transformed from one to four zones and how imbalance between supply and demand in a zone can lead to *"significantly higher prices"*.

Italy has also been the subject of more analyses. Fianu and coauthors analysed Italy with a focus on economic risk propagation between zones due to insufficient transmission capacity and uneven distribution of loads and productions [35]. De Siano stressed the connection between the spatial nature of electricity systems, zonal areas and merit order effects of power generators [36], and Wu and coauthors assessed market zone reconfiguration of Italy [37] using an evaluation framework developed for the purpose. They found improved market efficiency – albeit *"comparatively lower reduction in greenhouse gas [..] emissions"*.

Under Europe's zonal market design, all bidders within a zone face the same price regardless of their location. When grid expansion lags and intra-zonal congestion arises, these uniform prices cannot signal local scarcity or surplus, leading to both short- and long-term distortions. Bertsch et al. [38] use a decade-by-decade cap on transmission build-out in a pan-European, long-term power-system model to show that under the current market-design incompleteness and intra-zone grid under-investments (i) cause generation to be sited suboptimally, (ii)

creates regional imbalances, and (iii) escalate congestion - ultimately jeopardising policy and decarbonisation targets.

Höschle and coauthors [39], addressed the European electricity market and its zonal layout, pointing to national capacity mechanisms that *"undermine the process of harmonization and complicate efficient market coupling"* – in essence, distorting systems level signals.

In general, studies on market zones on Europe are dominating the scientific literature, but the literature also includes works on other areas. The National Electricity Market in Eastern Australia is composed of 16 market zones. In Ref. [40], Li et al. analysed PV and wind development in Australia and also point to the influence of capacity constraints between the market zones.

South Korea is a single-market zone country, but acknowledging potential social welfare impacts, Kwag and coauthors [41] investigated both zonal and nodal markets for the country, finding positive impacts of a zonal reconfiguration – but also potential conflicts of interest with the national TSO.

Navon and co-authors [42] investigated transmission congestion challenges related to the integration of distributed energy sources in an Israeli setting and found that there is a mismatch between scaling up PV and the transmission system, resulting in grid bottlenecks. Their analysis show that this can be relieved via optimal distribution of PV in the short term and coordinated expansion of grid and generation capacity in the long-term. They suggest moving from an unform price to a locational price signal such as a location-based tariff to support optimal distribution of VRES to avoid grid bottlenecks.

A possible move towards offshore energy hubs introduces a novel angle to the discussion, as such hubs have the potential of having generating or demand capacity comparable to that of a small nation or a market zone. This is for instance addressed by Kitzing & Gonzáles [43]. With several offshore energy hubs being considered in Northern Europe, these can have large effects on energy systems and markets alike – and how they are integrated into markets is quite relevant for the design and operation of energy systems beyond the offshore energy hubs.

Lund and Hvelplund discuss the neo-classical concept of free markets, arguing that for a market to be considered free, it requires many mutually independent suppliers and buyers, information on products and costs, and rational agents maximising profits and utility [44]. It could be added that free markets require that goods and services can be exchanged freely, and if this is not the case then it is either not a free market – or it is indeed more separate markets. In the concept of electricity markets, the same logic applies; either goods can be exchanged freely, or markets are indeed separate.

In this paper we address the composition of the German electricity market and how this affects the Danish energy systems' development. Denmark and Germany are selected as cases for several interconnected reasons. Denmark is moving towards an energy system with significant VRES and electrification of heating and transportation and is ahead of most other countries in this endeavour. Flexible electricity consumption such as from electric boilers and large-scale heat pumps in district heating have seen a large growth in Denmark while other flexible consumers such as electrolysers are emerging. Flexible consumers provide system-wide benefits by enabling the utilisation of excess electricity production from VRES, and their production is sensitive to price signals on the electricity market. At the same time, electricity prices in Denmark are affected by a 12 times larger neighbour (by population) whose internal market design has effects beyond own borders. Germany is also a case of a large single-price zone country where market imperfections even cause extensive redispatch in the form of curtailment outside the country. Finally, both Germany and Denmark are high-VRES countries, and both are country cases where the VRES-market nexus is more developed than in other countries.

The topic addressed in this paper is important for further VRES expansion and electrification efforts as market zone configurations affect stakeholder economy – and thus incentives in the energy transition.

Market design is a multifaceted issue which can be targeted from quantitative as well as qualitative angles. In the analyses presented in this article, it is addressed from both angles. In Section 2, an extensive literature review probes deeper into the journal literature in the field. Here we focus on previous analyses on zonal market designs, both analyses under existing designs and under alternative designs along with analyses of nodal designs. Section 3 presents an investigation of electricity price statistics with focus on price alignment and misalignment between adjoining market zones. This focuses on Denmark and the degree to which Denmark follows German and Norwegian prices. In Section 4, we address the energy system effects of varying price levels, taking advanced district energy plants as a case. Investigations of which heat production units are operating with different electricity price data years are presented. In Section 5 we look at special regulation, curtailment, and public perception. This provides a critical view of present curtailment practice – an effect derived from the market designs. Curtailment is one type of redispatch which can be employed to settle imbalances not settled by the day-ahead market. This is followed by an overview of market coupling mechanisms in the European Union (EU) in Section 6, providing the regulatory frame market designers must follow. Finally, conclusions are drawn.

The German-Danish market coupling is unique in the sense that that Denmark probably has the highest degree of VRES in its electricity supply while also aiming for the electrification of various parts of the energy sector – heating, transportation and industry. And at the same time is heavily influenced by a neighbour's electricity market design. While this situation is special and case-specific, the setup can also be seen as a frontrunner where other countries will experience similar issues as VRES and electrification efforts proceed. Experience with grid constraints, VRES and electrification efforts are also relevant in other places with large single market zones.

In the paper, a clear demarcation is being market splitting and zonal splitting, where these terms may otherwise in the literature be less consistently applied. For market splitting, we refer to the dynamic effect of two adjoining market zones diverging in electricity prices in a given hour due to the inability of the system to accommodate trades and resulting physical flows which would be effectuated if no constraints were there. In contrast to this, the term zonal split is applied when addressing changes in the static composition or design of markets – i.e., changes in the geographical expanse of individual market zones. Furthermore, market zones are called different things in different sources – market zones, price areas, price zones, bidding zones and probably more. In this article, the term market zone is used.

2. Zonal and nodal markets in the literature

There is an important literature on zonal and even nodal configurations of electricity markets. This section presents a review of work based on existing zonal configurations, alternative zonal configurations and lastly considerations of nodal markets.

2.1. Analyses with present zonal configuration

In Ref [45], Lindberg investigated power markets from a stakeholder perspective, comparing the situation in Scandinavia with several market zones to the German single-zone system. She found that in terms of energy transition and costs, the Scandinavian multi-zone system was more fitting whereas the German system formed a better basis for long-term trading. Lindberg found that *"Overall, the Nordic market is found to be better aligned with the energy transition in terms of flexibility, portfolio effects and system costs. The German market has considerably higher system costs but provides better conditions for long-term trading."*

Monforti-Ferrario and Blanco [46] see transmission grid congestion as a main issue in Europe, however taking Germany as an example, the authors were not able to find increasing emissions arising from the resulting redispatching. Redispatching may partially be seen as a

consequence of market zones not properly reflecting the physical reality of the electricity system, and in effect contributes to inefficiencies in price signalling.

In [47], Figueiredo and coauthors investigated wind power development in Denmark and potential congestion between the two market zones in Denmark with a focus on price convergence and market splitting. They found that wind power and congestion have a clear impact on market splitting. They found that the interconnection between the two zones of Denmark adequate, but also found that market splitting between Denmark and Germany should be avoided.

Cassetta et al. probed into convergence in consumer electricity prices across Europe finding a larger convergence among non-domestic consumers than within domestic consumers. The divergence among domestic consumers was primarily policy related. In addition, the authors sought to group the countries into convergence clubs, where both Denmark and Germany where divergent for household consumers while, e.g., the EU's Mediterranean member countries were in the same convergence club [48].

The European TSOs have conducted a comprehensive review of market zones in 2025 [49], where they have compared the present zonal configuration in European countries with alternative zonal configurations. They showed that splitting Germany–Luxembourg into five market zones gives a high economic value, ranging from €251 million to €339 million for the 2025 target year. One criticism of this review is that they do not reference any of the academic research on zonal configurations.

2.2. Analyses with alternative zonal configuration

Sweden is a sparsely populated country with diversity of energy sources in terms of availability and exploitation, ranging from fossil power stations, nuclear powers stations to wind and hydro power. In 2021, the Swedish electricity market was split into four market zone coming from a uniform national market zone. Comparing to Germany, it is clearly an interesting development seeing a large European country splitting its electricity market into more zones, however the focus of Lundin's work [50] was on the effect on wind power project development. Small developers were not affected by the zonal split while larger developers with a lesser localised focus were more affected in terms of where to establish wind power projects.

In Ref. [51] Sifnaios and coauthors used the Balmorel model (see Ref. [52]) to analyse a Scandinavian/German energy system transition with focus on the role of thermal energy storage in the energy system. In their analyses, they split Germany into four market zones, though this was done for bottleneck analyses and price formation only – and not as a stringent assessment of the zonal composition and potential market splitting. Their zonal split assumption indirectly shows that systems' analyses simply cannot be made on Germany on a single-zone market, as this does not reflect the physical reality.

Grohnheit & Sneum [53] analysed electricity prices in Scandinavia with a perspective on the effects of increasing penetrations of wind power typically offered at low or zero cost to the marginal price market – and as part of this also investigated the empirical effects on adjoining market zones. In general, they found a well-functioning system regarding market zones – though did not explicitly focus on the zonal composition.

In [54], Dobos and coauthors analysed alternative zonal configurations based on the European Union Agency for the Cooperation of Energy Regulators (ACER) for Germany. They found similar average prices across different zones after a zonal split, and did not find clear messages in favour of a zonal split based on their analyses of Germany alone.

Glynos and coauthors [55] also investigated alternative zonal configurations in Germany but with a focus on grid congestion, finding that despite zonal reconfigurations, congestion would persist. They concluded that market, RES development and grid development needed to be planned holistically.

Based on an analysis of European market zones and investment incentives, Grimm and coauthors stress the importance of a good designation of zones [56]. However, they also found that more market zones can reduce overall welfare. This is based on investment incentives to producers combined with considerations of competition being reduced inside smaller market zones. In an analysis of the Italian electricity market, the issue of market splitting was also addressed, and linking this discussion to the potential need for capacity payment in the future.

2.3. Zonal and nodal markets

Eicke & Schittekatte [57] mention the national starting points that have developed the European electricity market into a zonal market and contrast this zonal market to the nodal markets used in several other places including much of the United States. Electric Reliability Council of Texas (ERCOT) [58], which covers 75 % of the area of Texas and thus approx. 1.5 times the size of Germany, has in excess of 4000 "market zones" in contrast to Germany's single-zone market. Countries like Argentina, Chile, Mexico, New Zealand, Peru, Russia, and Singapore are also based on nodal market configurations [57]. In Ref. [59], Macedo and coauthors probe into the New Zealand nodal market, but face issues with data complexity and availability.

2013 Oggioni and Smeers [60] used a simple illustrative six-node model already in 2013 to analyse counter-trading and redispatch schemes in zonal markets, arguing for the need of a dedicated market in a European context to avoid large congestion-related costs, due to lack of sufficient signalling in the zonal market approach, which could be more adequately provided in a full nodal pricing scheme. Breuer and coauthors [61,62], developed an algorithm to evaluate a full nodal pricing approach across Europe, and the effects of splitting existing market zones based on grid-related indicators, such as grid congestion, rather than administrative borders. In 2015, Bucksteeg and coauthors considered a nodal market for Germany, but settling for analysing market splitting, finding a nodal market unrealistic at the time [63].

Ref. [64] analysed and compared wind power investment under zonal market pricing in a German single zone setting and a nodal market pricing setting using a German power-system model with detailed DC-flow constraints. The study found that nodal pricing results in investments with lower capacity factors, but more effective grid integration resulting in lowered system costs. They also highlighted the political challenges related to this as consumer prices increase slightly from the nodal pricing setting.

In [65], Lin and coauthors discussed nodal and zonal market configurations for Europe, analysing an alternative zonal configuration which could decrease congestion costs. Lin and coauthors also investigated how market zone reconfiguration in Europe could increase social welfare by reducing collusion on the electricity market [96].

Lété and coauthors [66] analysed market zones from the perspective of capacity expansion, finding that nodal pricing outperforms other pricing regimes in this aspect. They conducted their analyses using Central Western Europe as a case. Marinho and coauthors [67] suggested a zonal system based on nodes and analysing a central European system, they found that a candidate for reconfiguration of electricity market zones through a split would be in the Germany/Austria/Luxembourg area.

A motivating factor for nodal pricing is its ability to factor in local conditions including constraints and losses. In this respect, it shares some characteristics with the former renumeration scheme for cogeneration of heating and power (CHP) stations in Denmark, where renumeration was based on system value dependent on, e.g., voltage level of the transmission grid entry point [68] – though this was not based on dynamic prices but rather fixed in certain time slots. In their work, Eicke & Schittekatte investigated six potential reasons for the non-adoption of nodal pricing in Europe, but also argued against these [57], thus arriving at the conclusion to *"reconsider a nodal market"*.

In a report by FTI Consult from October 2023 on nodal or zonal

market design in Great Britain (GB), the authors found a potential of "*£28 billion to £51 billion of bill reductions in GB between 2025 and 2040 in a nodal market design, and £15 billion to £31 billion in a zonal market design*" [69].

3. Spot market electricity price alignment between Denmark, Germany, and Norway

This section demonstrates the price variation between Western Denmark (Market Zone DK1) and Germany (Market Zone DE) and Norway (Market Zone NO2) respectively. As shown in Fig. 1, Denmark has two market zones. Both of these are connected to the German Market Zone DE as well as the Swedish SW4 Market Zone through separate transmission lines. In addition, DK1 is connected to the Norwegian NO2 as well as to the Netherlands and to Great Britain. In general, Denmark is extremely well connected internationally, with more than 9 GW of capacity abroad (see Table 1)

For historical reasons DK1 and DK2 are non-synchronous areas with DK1 being synchronous with Central Europe and DK2 being synchronous with the rest of Scandinavia. DK2 was connected to SW4 in 1915 through an AC connection – nearly a century before DK1 and DK2 were connected in 2010 through a DC line.

A reason for the very strong connections in and out of Denmark is its role as a corridor between Norwegian and Swedish systems with significant hydro power and Central European systems dominated by thermal power production. After the significant wind power expansion in Denmark, this has also driven interconnection capacity expansion [14,15,79] at the same time of which the role of the interconnectors has changed.

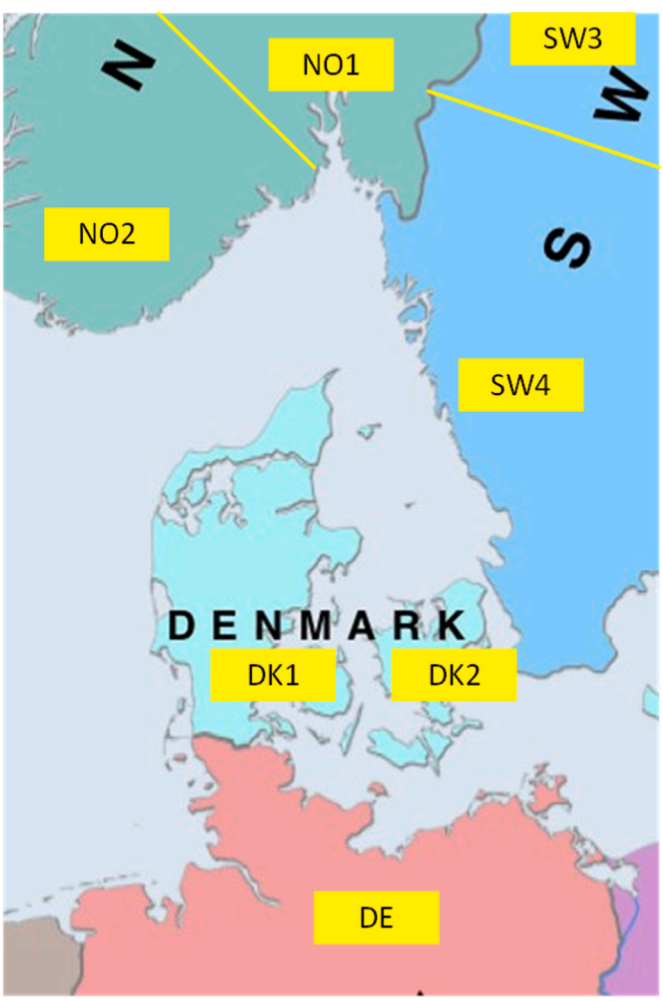

Fig. 1. Markets zones in Northern Europe/Southern Scandinavia. Deliniations between NO1 and NO2 in Norway and SW3 and SW4 in Sweden are indicative only.

Table 1
Connections to the Danish Market Zones DK1 and DK2. Note that the Viking Link cannot use its full capacity due to constraints elsewhere in the systems. Note also that most connections are made in more parallel systems, which have been supplemented over the years, renewed and upgraded. Hence, Commissioning year is the year of the first connection – and not the year of the full capacity listed. Lastly, note that while DK1 was connected to Germany already around 1929–1930, the area was still largely insular and main systems were only connected longitudinally in the Jutland peninsular in the mid-1950s. Based on data from Refs. [70–78].

Connection		Connection name	Capacity	Commissioning
DK1	NO2	Skagerrak	1700 MW	1976
DK1	SW4	Kontiskan	740 MW	1965
DK1	DE	(No name; normal AC aerial lines)	1780 MW	1929–1930
DK1	Great Britain	Viking Link	800/ 1400 MW	2023
DK1	Netherlands	COBRAcable	700 MW	2019
DK1	DK2	Storebælt	600 MW	2010
DK2	SW4	Øresundsforbindelsen	1700 MW	1915
DK2	DE	Kontek	600 MW	1995
DK2	DE	Kriegers Flak Combined Grid Solution	400 MW	2020

Showing data for 2023, the focus is on the level at the markets and whether prices in Denmark (DK1) align with or deviate from prices in two primary neighbouring countries – Norway and Germany. In general, when the market does split between Denmark and Germany, prices in Denmark are lower than in Germany, as shown in Fig. 2. For 64 % of the hours, prices are the same, for 35 % of the hours, prices are lower in Denmark and for 1 % of the hours prices are lower in Germany. Based on this alone it would be an overinterpretation stating that a further market split would lead to lower prices in Denmark, but the numbers do suggest, that Germany in general prop up prices in DK1. This aligns well with the previously mentioned analyses of Kristiansen [25].

Observing the connection to Norway in Fig. 3 gives a completely different picture. Here the price is only the same in 9.2 % of the hours of 2023, while Denmark is more expensive 62 % of the hours and cheaper 28.8 % of the hours. At 1700 MW, the connection from Western Denmark (DK1) to Norway (NO2) is very comparable to the 1780 MW connection between Western Denmark and Germany. For both numbers, domestic constraints are not considered, but it is apparent that despite comparable connection to Norway, Denmark and Norway are far more likely to split than Denmark and Germany.

Looking at the pie chart for 2023 in Fig. 4 shows that for the largest part of the year, DE and DK1 had spot market prices at a level higher than in NO2 (41 % of the hours), while in about one third (32 %) prices were different in all three market zones. Only during 4 % of the hours were prices identical from DE through DK1 to NO2. Without constraints, this would have been the dominant situation. In 62 % of the time, DK1 was at the same level as DE while this was only the case with NO2 in 10 % of the time. Fig. 4 thus supports that NO2 and DK1 split far more often than DK1 and DE – and in fact, if DK1 followed NO2 more, then prices would be lower. Clearly, if markets split due to actual congestion, then this is in line with the nature of the market – but if other factors trigger the split, then it is a potentially avoidable split.

Observing the pie charts from the three years shows an increase in hours where market splits occur such that all three differ and notably a decrease in hours where DK1 and DE are lower than NO2.

An observation can also been made domestically in Denmark where average annual spot markets prices in DK1 were lower than in DK2 for 15 consecutive years from 2006 to 2020 – but higher from 2021 to 2023. In 2023, spot market prices were identical in DK1 and DK2 65 % of the

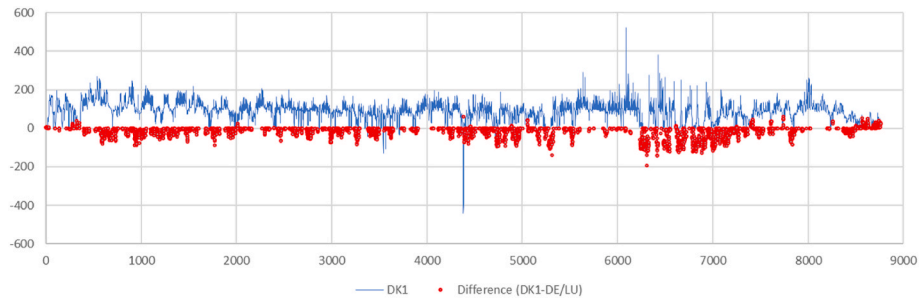

Fig. 2. Spot market prices in 2023 in the DK1 and DE/LU market zones. Red dots indicate the price difference when the market splits with negative numbers indicating a lower price in Denmark than in Germany/Luxemburg.

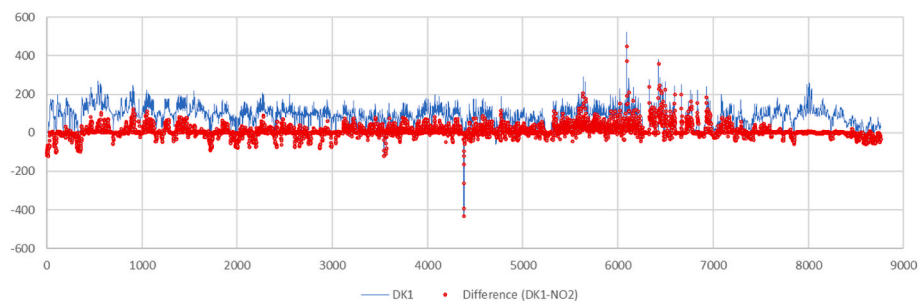

Fig. 3. Spot market prices in 2023 in the DK1 and NO2 market zones. Red dots indicate the price difference when the market splits with negative numbers indicating a lower price in Denmark than in Norway 2.

time – and lower in DK2 24 % of the time. A hypothesis could be that stronger international connections to non-Scandinavian countries in DK1 drive up spot market prices there.

4. Electricity price impacts on the energy system

Higher electricity prices benefit producers of electricity but not consumers of electricity. Entities that are both consuming and producing offer a more diverse picture. Clearly, the feasibility for electrification efforts within heating, industry, and transportation – including indirectly through PtX – is affected negatively by higher electricity prices. In this section, we look into the operation of a Danish district heating (DH) plant located in Ringkøbing in the DK1 market zone near the West Coast. Ringkøbing DH has previously been presented in the literature, with analyses of its solar field [80], power to heat [81], and uncertainties in operation and market performance [82].

Ringkøbing DH is equipped with an 8.9 MW_e natural gas engine, a 40 MW gas boiler, a 12 MW electric boiler, a 1.1 MW_e heat pump, two thermal storages of 250.3 MWh and 149.5 MWh respectively and 30000 m^2 of solar collectors tilted 30°. Thus, it can both take advantage of low and high electricity prices, and the storage provides opportunity to shift heat production temporally from heat demand.

For the analyses in this paper, it is relevant looking into which units that are optimal to run under which conditions. For this purposes, net heat production price (NHPP) curves are made. These NHPP curves provide the short-term marginal heat production cost at different electricity prices. The cost includes spot market prices of electricity, applicable taxes, grid fees, variable operation and maintenance – but does not, for instance include fixed operation and maintenance or investment costs or any profit. It is thus the price curves the operator will observe when deciding on the operation in a given hour.

Fig. 2 shows the NHPP curves for April 2022. The curves vary over the year, however for these analyses fixed curves have been used. This thus disregards that the natural gas price varies (thus affecting natural gas boiler and the gas engine) and that the temperature varies which affects the coefficient of performance (COP) of the heat pump. The curves are calculated using the energyPRO energy systems simulation model (see Ref. [83]).

Solar collectors and the gas boiler are not affected by the electricity price, but the other NHPP curves are either ascending or descending, giving intersection points and thus equilibrium prices where the cost of generating heat from electric boilers and heat pumps are the same and where the cost of generating heat from the gas engine or the heat pump is the same.

Electricity prices must be negative for the electric boiler to be the most favourable option (below −75.4 DKK/MWh (−10.1 EUR/MWh)) while prices higher than 568 DKK/MWh (75.7 EUR/MWh) make the gas engine more favourable than the heat pump. The solar collector is typically the most favourable option (not for very low prices), and the gas boiler is never, hence these units are used if available (solar) or pertinent (gas boiler).

Clearly, when making bids on the electricity market, producers and consumers do not know the clearing price of the individual hours beforehand, thus bidding price intervals cannot be directly compared to clearing prices, however bidders on the electricity market have access to tools which based on available data – including meteorological data – provide forecasts for the electricity market. Such tools are essential for all bidders, and here we assume that bidders act on forecasted price signals that largely reflect the final clearing prices.

Observing electricity spot market prices for nearly five full years, the different units are optimal in varying degrees from year to year, as shown in Fig. 6. For the first years, the heat pump was clearly the most favourable option, while in the latter years, the gas engine became more favourable. Electric boilers are only modestly favourable – no more than 53 h per year for the shown years as there is only a limited number of hours with negative prices (negative prices have been allowed since 2009 [84]).

The higher prices the latter years – and thus the decreasing feasibility of heat pumps and electric boilers compared to the gas engine – is a fingerprint of first a rapid post-COVID19 recovery followed by supply crises and the phasing out of Russian natural gas after the invasion of Ukraine in 2022. While it should be noted again that the chart is made

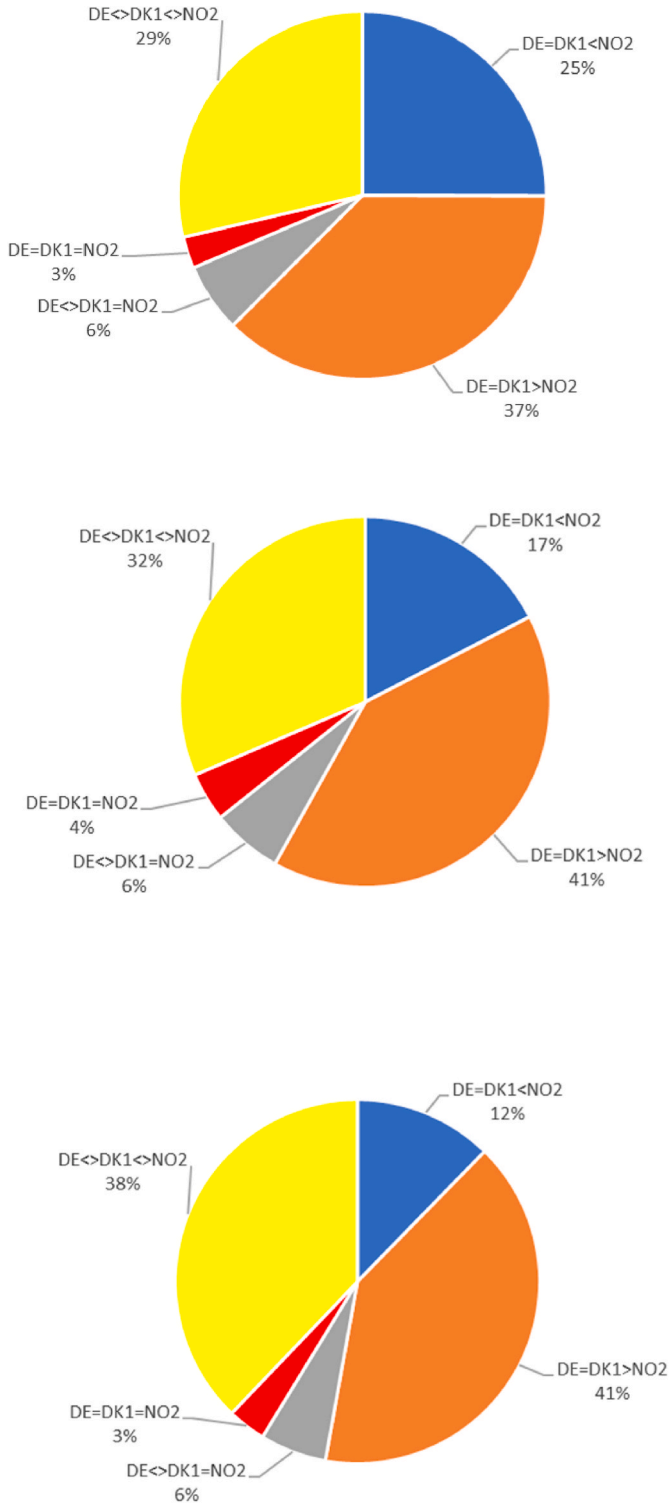

Fig. 4. Alignment between spot market prices in the DK1, NO2 and DE Market Zones. Based on number of hours using data from 2022 (top), 2023 (centre), and 2024 (bottom).

based on a fixed April 2022 natural gas price, it also shows that under normal circumstances, heat pumps are the preferred option. Thus, while high electricity prices – as, e.g., caused by grid congestion in Germany without the splitting up of the market – are favourable for the operation of the gas engine, it is not favourable for the heat pump foreseen to play a pivotal role in sustainable heating [9–12].

Lower prices clearly also affect power production. Not only VRES-based producers (wind and PV) but also, e.g., biomass or biogas-based CHP and other thermal power production units. Future systems can thus need some market adaptions to ensure feasibility of these units – possibly in the shape of a capacity market. As also deliberated in Ref. [85], a new market structure may also be needed in the future. A situation acerbated by increasing amounts of power being bid at a short-term marginal production price of zero.

For the above analyses, only the spot market has been considered. CHP units, electric boilers and heat pumps also have the opportunity to act on short term markets – frequency containment reserve markets for instance. However, the traded quantities on the spot market constitute the majority of the total trade.

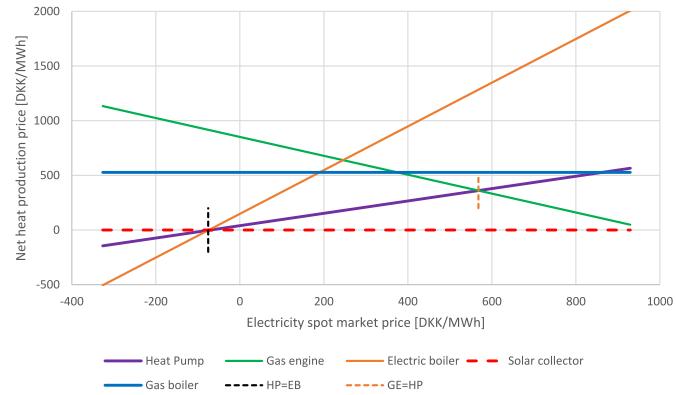

Fig. 5. Net heat production price curves for Ringkøbing DH with temperature and natural gas price data for April 2022.

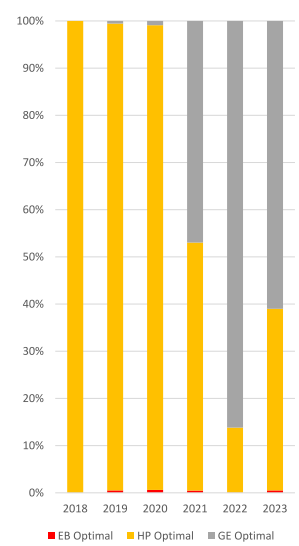

Fig. 6. Number of hours where the three units operating on the spot market are optimal for producing heating over a five-year period. EB equals electric boiler, HP equals heat pumps, and GE equals gas engine. Based on the net production price curves in Fig. 5.

5. Special regulation and public perception

The situation with lacking zonal market split has caused dissatisfaction in Northern Germany [86] as well as across the border in Denmark where downward regulation is being purchased from Germany. Curtailment of wind power – as an effect of a need for downward regulation – has caused some level of public puzzlement in Denmark – despite positive prices in the Day-ahead market – to the extent that the Danish TSO has had to make a dedicated webpage explaining why it

occurs [87]. This has been one of the aftermaths of the Russian war on Ukraine and the subsequent phase-out of Russian natural gas and other fossil supplies to Europe. Electricity prices have risen – and, e.g., Danish consumers have been unable to comprehend how high prices could coexist with wind power curtailment – and basically why Denmark as a nation has been unable to harvest benefits from the nation's long-standing commitment to wind power during a time of crisis.

This curtailment of Danish wind power constituted 7.3 % of the total production in 2021 [87], and while some consumers concern that the consumer price is affected, the TSO argues with basis in the structure of the spot market and that prices were settled taking the wind power production into consideration, that prices are largely unaffected [87]. However, they still acknowledge the poor signal represented by stopped wind turbines.

Special regulation is a model for countertrading of electricity exchanges across the German-Danish border. Special regulation is ordered and paid by TenneT (the German TSO with land borders to Denmark) to address substantial imbalances due to excess production in Northern Germany. These imbalances are primarily attributed to bottlenecks within the German transmission system, which hinder the efficient transportation of electricity from the point of generation to areas of high demand in the south.

Under German regulatory policies, the curtailment of wind power production within Germany is considered a measure of last resort. Consequently, the bottleneck challenges become apparent in the network's struggle to accommodate the volume of imports from DK1, that are agreed upon in the day-ahead market. Given that these bottlenecks remain unaddressed in the day-ahead market framework, where this prompts countertrading to balance the imbalances, this is often resulting in the curtailment of wind power in Denmark as evident in Table 2. Moreover, this situation prompts a redispatch of German thermal power plants located outside the congested areas, as a strategic response to maintain grid balance.

Table 2 provides an overview of the activated volumes of downward regulation for special regulation and the split between delivered downward regulation from Danish participants in DK1 in the period from 2018 to 2020, with 2020 marking a record year. Special upward regulation is also traded, however only in rare occasions. Downward regulation is the predominant order in terms of traded volumes as it alleviates the described issues surrounding instances with excess production imbalances in Northern Germany, due to bottleneck challenges. The table shows that a total of 3 TWh of special downward regulation occurred in 2020, corresponding to approximately 11 % of the total annual electricity production in Denmark in the period. Curtailed wind power delivered 47 % of the special down regulation in 2020, corresponding to 1.4 TWh of curtailed wind power; approximately 9 % of the total wind power production in Denmark in the period. Electric boilers and other consumption types contributed with approximately 17 %, while thermal power plants were responsible for the remaining 36 %.

From 2023, the model for special regulation governing countertrading across the German-Danish border has been integrated into the intra-day market framework. This development opens opportunities for other Nordic market participants, including Norwegian hydro power plants, to actively participate and bid in the special downward regulation. This integration represents a notable expansion of the market dynamics, allowing for a more diverse and broader participation in the special regulation across the region, however it does not solve the underlying issues causing the inefficient market allocation of resources.

Table 2
Activated volumes of downward regulation for special regulation (GWh), split by source (%). Source of data [88–90]:

	2018	2019	2020
Special downward regulation (GWh)	1134	1334	3033
Delivered by wind power	26 %	32 %	47 %
Delivered by consumption	21 %	22 %	17 %
Delivered by other production	53 %	46 %	36 %

6. Coupling of electricity markets in Europe and investor impacts

The European Union Agency for the Cooperation of Energy Regulators (ACER) [91] was established in 2011 and its primary role is to ensure cross-zonal trading opportunities in Europe as *"Maximising cross-zonal trading opportunities is a core element to ensure an efficient internal electricity market. It is also important for meeting a number of future challenges in Europe's energy trajectory, including the decarbonization targets."*. As part of this, they have institutional oversight of the minimum level of available cross-zonal capacity, where TSOs must ensure that 70 % of the transmission capacity is available for trade across market zones. This 70 % requirement came into effect in 2020 and applies both between countries with distinct zones and within countries with more zones [92]. TSOs can use different handles to ensure meeting this requirement, including grid expansion, redispatching & countertrading and redesign of market zones [92]. Every three years, ACER also assess the European market zone configuration [93] with input from the European Network of Transmission System Operators (ENTSO-E).

Euphemia [94] establishes the more exact price and market coupling mechanisms between different market zones. In essence, the primary objective is to enable the selling and buying across market zone boundaries – restricted only by the transmission capacity between zones. Sales and purchase bids are assessed based on grid restrictions with the objective function of maximising the social welfare. This social welfare is defined as the sum of three components: Consumer surplus, Producer surplus and Congestion rent.

Consumer surplus is defined as the aggregation of executed purchasing volumes multiplied by the difference between the respective purchasing bid levels and the clearing price in the individual hour, where the market clearing price is the level that would be determined if no transmission constraints existed between adjoining zones. In a similar way, producer surplus is determined based on executed purchasing bids, bid levels, and clearing price in the individual hour. The congestion rent is calculated as the product of the flow through the interconnector and the difference in clearing prices between the adjoining zones. Hence it follows, if the market is not split in an individual hour, the congestion rent is zero.

The computation of the interconnected systems seeks the flows and clearing prices that will optimise social welfare, and the computational complexity is aggravated by a series of constraints including complex bids (e.g., block bids), ramping, losses, line tariffs and more. A particular peculiarity is the avoidance of non-intuitive flows (also known as *adverse flows*), where optimal solutions may result in flows from high-price to low-price zones. The high complexity results in a mathematical problem requiring Mixed-Integer Quadratic Programming when implemented full-scale in the European connected markets, and development has been a long-standing effort [94].

A model like Balmorel also seeks to optimise this social welfare with the objective of finding flows and clearing prices in the different market zones for planning purposes, thus the more advanced market features are not relevant.

Given the asymmetry in the geographical distribution of electricity supply and demand, splitting Germany into more market zones would lower the prices in North Germany, sending an important signal to investors in, e.g., PtX-plants, to place the plant close to offshore wind. However, by having Germany as a one market zone, the opposite signal is sent to investors. The single market zone means of course that you can buy electricity in North Germany and South Germany to the same price, however, in the areas (like North Germany) with a significant VRES and less households the transmission grid tariff is higher.

As an example, industrial customers are paying 3.35 €¢/kWh in Schleswig-Holstein at the northern end of Germany and 2.48 €¢/kWh in

Bayern at the southern end in grid costs. Commercial customers are paying 6.75 €¢/kWh in Schleswig-Holstein and 5.28 €¢/kWh in Bayern [95]. So as an investor in PtX you get a price signal to place your PtX in South Germany, which will create significant difficulties for the German transmission grid.

The outlined distortions in price signalling in the German market-zone can in effect result in suboptimal decision making that, instead of addressing grid-related bottleneck challenges, actually amplify these.

7. Conclusions

This article has explored the field of the electricity market zonal configuration with a focus on the situation between Denmark and Germany, where Denmark has two zones – and Germany despite its large expanse and bottlenecks between south and north is composed of only one market zone, giving cause to market distortion also outside of Germany.

Based on an extensive literature review, the article has outlined the issues and circumstances related to this situation – and for instance found a large body of work on both zonal configurations as well as nodal considerations.

There is some alignment between electricity prices in Denmark and Germany and some between Denmark and Norway, however market splits typically cause prices in Denmark to become lower than in Germany – indicating that a truer German market would generate lower prices in Denmark. The effects of the prices have been investigated based on a Danish district heating plant, where lower prices will clearly impact heat pump and electric boiler operation positively. Moreover, similar effects can be expected for future market participants, such as operators of electrolysers or PtX. Clearly, remuneration for power generation will be affected negatively, though this has not been a primary topic for this article.

While the quantitative analyses are specific for this case, the situation is not. More large countries in e.g. Europe form single market zones thus potentially impacting neighbouring countries with electricity market price signals that may be distorted by the market zone configuration.

Public perception is important for the energy transition and in the Danish context acceptance of wind power is important – and special regulations caused by grid congestion across the border in Germany and subsequent curtailment of wind power in Denmark, has generated public opposition. A situation aggravated through complex market constructs in the European electricity market. EU electricity market regulation does say that zonal design should be considered, however there are clearly elements still left unaddressed.

Research point towards a potential adoption of nodal markets, which would clearly provide incentives for the correct placement of new demand and production facilities – also factoring in bottlenecks in present transmission systems.

Policy implications for the present work are ambiguous, as different stakeholders are affected differently. While at odds with ACER stipulations, Germany must value a uniform price higher than cost-reflecting prices. This favours demands in the south of the country as well as production in the north of the country. In Denmark, it favours production facilities while at the expense of consumers as well as electrification efforts. These cannot benefit from the low prices that zonal reconfiguration of Germany combined with market splits would lead to in high VRES hours. Policy is a matter of appropriately balancing this, but as this work focusses on district energy plants and the economic feasibility of electrification efforts, alignment with ACER stipulations is preferable.

In our analyses we have focused on demonstrating the problem and the implication for electricity price levels for DH plants. The quantitative analyses have therefore been empirical analyses of historical prices. In the article, we have not quantitatively analysed the effects of a zonal split in Germany. This is a clear future research direction to be addressed. Moreover, consequences of the current configurations and effects of a zonal split in Germany with focus on emerging market participants such as PtX or electrolysers is an area to be addressed in future research, as the additional electricity demands from such actors are poised to be astronomical in the transition towards a net-zero EU.

CRediT authorship contribution statement

P.A. Østergaard: Conceptualization, Methodology, Project administration, Writing – original draft, Funding acquisition. **A.N. Andersen:** Conceptualization, Methodology, Writing – original draft, Funding acquisition. **F.D. Nielsen:** Writing – original draft, preparation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The work in this paper has been supported by the Offshoore Energy Hubs project with the support of EUDP 64022-1011. A preliminary version of the work was presented at the SDEWES conference in Viña del Mar, Chile, January 14–17, 2024.

Nomenclature

AC	Alternating current
ACER	European Union Agency for the Cooperation of Energy Regulators
CFD	Contracts for Difference
DC	Direct current
DH	District heating
EU	European Union
LCOE	Levelised Cost of Energy
NHPP	Net heat production price
PtX	Power to x
PV	Photo voltaics
RES	Renewable energy sources
VRES	Variable renewable energy sources
TSO	Transmission system operator

Data availability

Data will be made available on request.

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